Data Selection: 311 Service Request Forecasting and Drift Detection

Himanshu Parida
UIN: 237007758

Overview

The project ingests non-emergency "311" civic service requests (potholes, noise complaints, heat outages, missed trash pickup, etc.) as a live, continuously updating stream and uses it to (1) forecast short-horizon request volume by area and type, and (2) detect and respond to real distributional drift in a production-style, self-monitoring ML pipeline. Unlike a static Kaggle dataset, 311 data comes from live civic infrastructure that nearly every resident interacts with, and it exhibits real, dateable drift events (storms, policy changes, seasonal cycles) that can be tied to concrete causes rather than injected synthetically. It also suits a one-semester scope as the historical archive is immediately usable for offline development and the live feed needs no special access or infrastructure to poll, so most of the project's time can go toward the modeling and monitoring pipeline rather than data acquisition.

Data Source

- **Primary source:** NYC Open Data (Socrata platform), dataset *"311 Service Requests from 2010 to Present"* (Socrata UID `erm2-nwe9`).
- **Live JSON endpoint:** `https://data.cityofnewyork.us/resource/erm2-nwe9.json`
- **Fields:** request ID, created/closed/due timestamps, responding agency, complaint type/descriptor, borough/zip/address, latitude/longitude, status, resolution text, filing channel.
- **Coverage and cadence:** records since 2010 for historical training; the live feed is updated as new complaints are filed, supporting near-real-time polling for the streaming component.

Critical Assessment

- **Reporting bias, not incident ground truth.** 311 volume reflects who chooses to report, not the true rate of underlying issues. Higher-income, English-speaking, or app-savvy neighborhoods tend to over-report relative to actual conditions, while others under-report due to distrust issues, language barriers, or lack of awareness. Forecasts trained on this data risk encoding and reinforcing existing inequities in city resource allocation if used naively.
- **Taxonomy drift vs. real-world drift.** Complaint-type categories and agency assignments are occasionally renamed, merged, or reclassified by the city. A shift in category frequency may reflect a labeling change rather than genuine behavioral or environmental change, so drift detection must be able to distinguish schema drift from substantive drift.
- **Data quality.** `closedDate` and `resolutionDescription` fields are filled in asynchronously and inconsistently across agencies, and geocoding (lat/long) is occasionally missing or imprecise, which affects feature quality for the most recent, still-open requests.
- **Class imbalance and geographic sparsity.** Rare complaint types in low-density areas produce noisy, low-count series that are hard to forecast reliably and can dominate error metrics if not weighted carefully.
- **API is polled, not pushed.** The Socrata endpoint is a REST API queried on a schedule, not a true push-based stream (e.g., a message broker at the source). The project simulates streaming via periodic polling into Kafka; this is a reasonable and common real-world pattern but should be described accurately rather than as a native stream.

Mitigation Strategy

Aggregate forecasts to the community-district level rather than exact addresses; monitor category-level schema changes separately from volume-based drift statistics; use a time-based (not random) train/test split that respects known structural breaks (e.g., COVID-era reporting shifts); and explicitly log when the drift monitor fires due to a labeled external event (storm, policy change) versus an unexplained shift, to keep the causal narrative honest and auditable.

Conclusion

The NYC 311 feed offers a data source that is public, free, live, and rich enough for meaningful feature engineering. Also, it surfaces real fairness, data-quality, and drift-attribution challenges that a synthetic dataset would not. These limitations are treated as first-class design constraints for the monitoring and retraining pipeline, not incidental noise to be ignored.